CT-KAE candidates on Kuramoto–Sivashinsky

1 November 2026

Summary

The two transonic CT-KAE candidates (the reference and back001) were trained on KS with the transonic recipe and run through every KS experiment of the paper. The paper's KAE, U-Net and SDA are the baselines.

- **Forecasting fails beyond one training horizon.** The CT-KAEs have the best encode–decode accuracy of the four models (0.007), but their error grows ten-fold every ten steps. At $0.1\,T_L$ it is 0.58–0.64, against 0.024 for the paper's KAE and 0.0003 for the U-Net. Skill is lost at 0.3–0.4 T_L (paper's KAE 2.4, U-Net 9.3).
- **DA is close to the paper's KAE when the first observation is close to t_0.** The canonical analysis is 0.025–0.028 (paper's KAE 0.0044, SDA 0.009, U-Net 0.097). The CT-KAEs match the paper's KAE with ≥ 13 observation times. They are the best of all five models with 25–50% sensor coverage and many observation times.
- **DA is worse than the paper's KAE** when the first observation is later than 0.1 t.u., with noise (3–6×), and in long windows.
- **Of the transonic readouts, only the sharpened symmetry average is admissible on KS.** It halves the 50-frame error, to just under persistence. LPM fails.
- **back001 vs the reference:** back001 has a slightly longer forecast skill, within seed spread; DA is the same.
- **The cause is the training horizon.** The 8-step rollout (0.8 t.u.) of the transonic recipe is 3.5% of a Lyapunov time on KS, and the error leaves the encode–decode floor there. A KS recipe with a longer rollout would test whether the architecture or the recipe limits the forecast.

Setup

- **Models.** The two transonic candidates, the reference and back001 (the reference plus backward latent consistency at weight 0.01), trained on KS with the transonic recipe: latent dimension 256, widths 64/128/256, 128 damped oscillators (exact $e^{\tau A}$), no fixed point, reconstruction + 8-lead prediction + latent consistency + amplitude terms, 20k steps, batch 16, AdamW 3×10^{-4}, final checkpoint, 3 seeds each. KS-specific changes: 1-D convolutions with circular padding, one input frame (KS is Markovian; the paper's KS models also encode one frame), no physical parameter, $\Delta t = 0.1$ (the transonic step). Training: `train.nc` (80 trajectories); `val.nc` only logged. About 8 minutes per run on one RTX 4090.
- **Baselines.** The paper's KS KAE, U-Net and SDA. Their numbers are the paper's stored results on the same seeded problems. The paper's KAE cannot be re-run: with today's code its encode–decode error is 0.135 instead of the 0.009 recorded in August (identical architecture, strict checkpoint load; the forward pass probably changed in the September refactor).
- **Protocol.** The paper's KS code, unmodified, with the CT-KAE in the KAE slot (`audit_report/scripts/ks_ctkae/ks_driver.py`). Two adaptations, both decided on validation data: (i) the 4D-Var control is the latent normalised per dimension by its training-split standard deviation (an exact change of variables: the CT-KAE latents have a scale of ∼350, the paper's KAE ∼1, for which the learning-rate grid was built); (ii) the initialisation grid adds the harness's informed first guess $e^{-A\tau_1}E(y_1)$ to {random, zero}. Validation selected the informed first guess and learning rate 0.03 for both candidates (validation error 0.231 and 0.226; the paper's KAE 0.189 with zero start); from zero, the CT-KAE does not leave the latent origin's basin (validation error ≈ 1). DA results are for seed 0.

Forecasting

Table 1: Free-running forecast from the true state (`da_test_long`, 32 trajectories, $T_L = 22.6$ t.u.): relative L_2 error at leads in Lyapunov times, the skill horizon (error reaching 90% of the no-skill level) and the encode–decode error at t_0. CT-KAE: mean ± SE over 3 seeds.

	$0.1\,T_L$	$0.5\,T_L$	$1\,T_L$	$5\,T_L$	skill horizon (T_L)	round trip
U-Net	0.0003	0.0019	0.0038	0.4961	9.29	–
Paper KAE	0.0242	0.7443	1.1824	1.3824	2.42	0.0171
Reference	0.643±0.014	1.584±0.132	1.816±0.102	1.393±0.087	0.28±0.04	0.0069
back001	0.575±0.026	1.407±0.165	1.721±0.070	1.428±0.100	0.38±0.09	0.0077

The CT-KAEs are the most accurate models of the four over a few steps and the least accurate beyond one training horizon. Their encode–decode error (0.007–0.008) is below the paper's KAE (0.017 in this file, 0.009 in the paper's AE audit), and their one-step error on `test.nc` is 0.008. But the error grows ten-fold every ten steps (0.089 at 1 t.u., 0.51 at 2.2 t.u.; seed 0 reference), so at $0.1\,T_L$ it is already 0.58–0.64 against 0.024 for the paper's KAE and 0.0003 for the U-Net, and skill is lost at 0.28–0.38 T_L (paper KAE 2.4, U-Net 9.3). This growth is model error, not chaos: the learned generators are strictly stable (largest real part -0.0075 to -0.0088 over six runs; the leading Lyapunov exponent is $+0.044$). Training rolls out 8 steps (0.8 t.u.), which is roughly where the error leaves the encode–decode floor. back001 is ahead at short leads (0.575 vs 0.643 at $0.1\,T_L$; horizon 0.38 vs 0.28 T_L), with seed spread of the same order.

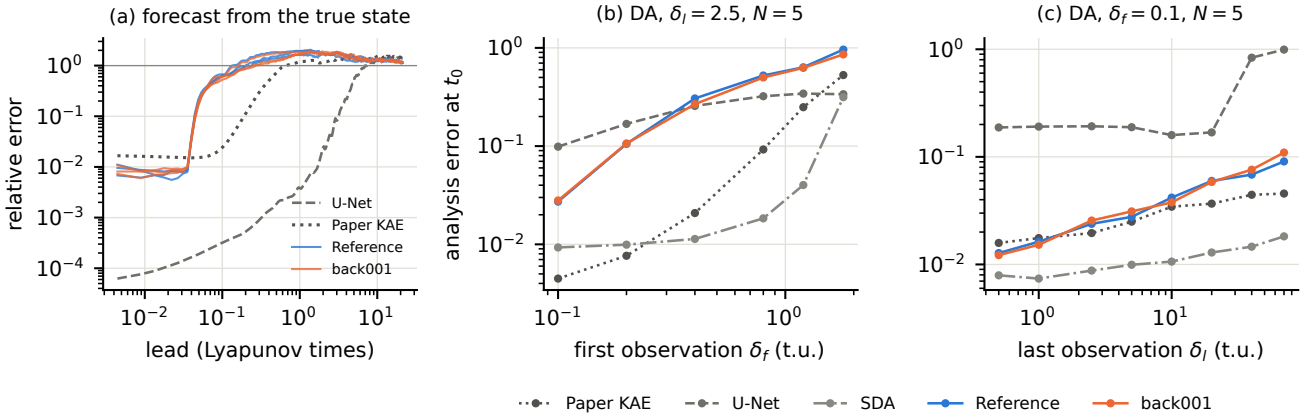


Figure 1: (a) Forecast error from the true state (CT-KAE: 3 seeds each). (b, c) Analysis error of 4D-Var against the lead of the first and of the last observation (48 and 24 problems; CT-KAE: seed 0).

Data assimilation

When the first observation is one step after t_0 (the paper's campaign), the CT-KAEs are DA models of the same class as the paper's KAE, ahead of the U-Net except at noise $\sigma \geq 0.2$, and ahead of SDA in places:

- **Canonical window** ($N = 5$): 0.025–0.028, against 0.0044 for the paper's KAE, 0.009 for SDA and 0.097 for the U-Net.
- **More observation times close this gap.** With 8 times the error is 0.009, and with 13 or 16 it is 0.006, level with the paper's KAE (0.0055–0.0060).
- **Sparse sensors with many times** are the CT-KAEs' best regime. At 25% coverage with 8 or 13 times they are the best of the five (0.014–0.019, against 0.070 for the paper's KAE and 0.018–0.021 for SDA). At 50% coverage with 13 times they are also best (0.008 against 0.010 for SDA).
- **Short windows** ($\delta_l \leq 1$ t.u., 1000 iterations): 0.012–0.016, slightly better than the paper's KAE (0.016–0.018).

Table 2: Analysis error at t_0 (relative L_2, mean over 48 problems; 24 for δ_l) of every sweep of the paper, on the same problems. Defaults: $\delta_f = 0.1$, $\delta_l = 2.5$ t.u., $N = 5$, full coverage, no noise; 8000 iterations (1000 for δ_l). Best per row in bold. N is the realised number of observation times (rounding to the frame grid drops duplicates).

	Paper KAE	U-Net	SDA	Reference	back001
first observation δ_f (t.u.)					
0.1	**0.0045**	0.0987	0.0093	0.0272	0.0279
0.2	**0.0076**	0.1684	0.0099	0.1054	0.1063
0.4	0.0208	0.2565	**0.0113**	0.3055	0.2676
0.8	0.0924	0.3220	**0.0184**	0.5231	0.4995
1.2	0.2489	0.3424	**0.0402**	0.6341	0.6262
1.8	0.5288	0.3375	**0.3148**	0.9604	0.8581
last observation δ_l (t.u.)					
0.5	0.0158	0.1877	**0.0079**	0.0128	0.0122
1	0.0176	0.1911	**0.0074**	0.0163	0.0152
2.5	0.0196	0.1924	**0.0088**	0.0238	0.0256
5	0.0250	0.1886	**0.0099**	0.0277	0.0311
10	0.0344	0.1593	**0.0106**	0.0418	0.0378
20	0.0367	0.1688	**0.0129**	0.0597	0.0584
40	0.0444	0.8355	**0.0146**	0.0683	0.0760
70	0.0456	0.9930	**0.0182**	0.0906	0.1097
2.5 (8000 it.)	**0.0041**	0.0970	0.0088	0.0262	0.0302
observation times N					
2	0.0422	0.0873	**0.0181**	0.0308	0.0294
3	**0.0075**	0.0931	0.0133	0.0339	0.0328
5	**0.0044**	0.0965	0.0091	0.0248	0.0268
8	**0.0049**	0.0994	0.0074	0.0091	0.0092
13	**0.0055**	0.1004	0.7377	0.0060	0.0058
16	0.0060	0.1006	–	**0.0059**	0.0059
noise std (Gaussian)					
0.0	**0.0046**	0.0960	0.0098	0.0377	0.0381
0.01	0.0076	0.0953	**0.0038**	0.0421	0.0456
0.05	0.0343	0.1748	**0.0135**	0.1139	0.1234
0.1	0.0739	0.2587	**0.0242**	0.2499	0.2651
0.2	0.1414	0.3780	**0.0417**	0.5923	0.6179
0.4	0.2383	0.5831	**0.0770**	0.9717	0.9657
coverage, N					
100%, 3	**0.0079**	0.0921	0.0139	0.0377	0.0355
100%, 5	**0.0042**	0.0952	0.0094	0.0252	0.0273
100%, 8	**0.0049**	0.0949	0.0076	0.0097	0.0096
100%, 13	**0.0053**	0.0952	–	0.0061	0.0062
50%, 3	**0.0139**	0.1120	0.0212	0.0417	0.0411
50%, 5	**0.0123**	0.1084	0.0147	0.0287	0.0277
50%, 8	0.0131	0.1073	0.0118	**0.0114**	0.0123
50%, 13	0.0142	0.1089	0.0098	0.0088	**0.0082**
25%, 3	0.0745	0.3238	**0.0346**	0.0403	0.0414
25%, 5	0.0704	0.1690	**0.0257**	0.0295	0.0275
25%, 8	0.0697	0.1602	0.0213	0.0185	**0.0170**
25%, 13	0.0698	0.1615	0.0180	0.0146	**0.0136**
10%, 3	0.5867	0.9710	**0.2632**	0.5154	0.5194
10%, 5	0.5867	1.0289	**0.2213**	0.5045	0.5086
10%, 8	0.5784	1.0260	**0.2093**	0.4922	0.4947
10%, 13	0.5761	1.0151	**0.2095**	0.4896	0.4871
single observation at τ					
0.1	0.0416	0.1084	**0.0171**	0.0228	0.0226
0.2	0.0898	0.1617	**0.0173**	0.0738	0.0736
0.3	0.1413	0.2324	**0.0177**	0.1544	0.1477
0.5	0.2554	0.2657	**0.0185**	0.3563	0.3230
0.8	0.4310	0.2853	**0.0202**	0.5322	0.4934
1.3	0.6632	0.3528	**0.0285**	0.5899	0.5985
2.1	0.9215	0.3709	**0.1422**	1.0238	0.8876

- **The first observation must be close to t_0.** The error grows to 0.11 at $\delta_f = 0.2$ and 0.27–0.31 at $\delta_f = 0.4$ (paper's KAE 0.008 and 0.021). This follows from the forecast error: inverting the model over δ_f costs what forecasting over δ_f costs. The main repository's $\delta_f = 0.9$ campaign (Table 3) shows the same: about 0.5 at $N = 5$, against 0.12 for the paper's KAE and 0.02 for SDA.
- **Noise is the weakest regime**: 3–6× the paper's KAE at every level ($\sigma = 0.05$: 0.11–0.12 against 0.034; $\sigma = 0.4$: 0.90–0.97 against 0.24–0.26).
- **Long windows**: on `da_test_long` the error rises from 0.024–0.026 at $0.11\,T_L$ to 0.09–0.13 at $17.7\,T_L$ (paper's KAE 0.018 to 0.073). In the δ_l sweep it rises from 0.024–0.026 at 2.5 t.u. to 0.09–0.11 at 70 t.u. (paper's KAE 0.046, SDA 0.018).
- **Forecast from the analysis** (Table 5): the analysis is good (0.036–0.037, paper's KAE 0.013), but the forecast from it inherits the propagator. Skill is lost at 0.34–0.40 T_L, against 0.53 for the paper's KAE and 10.9 for the U-Net. The U-Net's analysis error (0.147) is in modes that decay within a few steps; its forecast error at $0.1\,T_L$ is 10^{-4}.

The two candidates are within seed-level differences of each other throughout. back001 is not reliably better in DA.

Table 3: The same sweeps with the first observation at $\delta_f = 0.9$ t.u. (the main repository's `da_results_geometry_df9` campaign, the current default of `exp_geometry`; same schedules frame for frame). Defaults: $\delta_l = 2.5$ t.u., $N = 5$, 8000 iterations (1000 for δ_l).

	Paper KAE	U-Net	SDA	Reference	back001
last observation δ_l (t.u.)					
1.3	0.2752	0.4230	**0.0146**	0.5196	0.4841
2.5	0.2663	0.4576	**0.0209**	0.4928	0.4700
5	0.3016	0.4452	**0.0216**	0.5181	0.4610
10	0.2901	0.4291	**0.0219**	0.5773	0.5477
20	0.2822	0.3565	**0.0237**	0.5686	0.5306
40	0.3529	0.8788	**0.0243**	0.8130	0.7111
70	0.3833	0.9900	**0.0278**	0.7022	0.7050
2.5 (8000 it.)	0.1234	0.3427	**0.0210**	0.4915	0.4781
observation times N					
2	0.3928	0.3189	**0.0229**	0.5644	0.5157
3	0.1934	0.3245	**0.0225**	0.5418	0.5037
5	0.1196	0.3293	**0.0215**	0.5104	0.4863
9	0.1225	0.3318	**0.0190**	0.3847	0.3644
13	0.1259	0.3333	**0.0184**	0.3565	0.3426
17	**0.1376**	0.3319	–	0.3563	0.3414
noise std (Gaussian)					
0.0	0.1293	0.3671	**0.0214**	0.5184	0.5099
0.01	0.1532	0.3695	**0.0120**	0.5386	0.5351
0.05	0.4306	0.4962	**0.0257**	0.7829	0.8038
0.1	0.6060	0.7886	**0.0389**	1.1975	1.2139
0.2	0.6632	1.5046	**0.0617**	1.6678	1.6606
0.4	0.7259	2.8842	**0.1027**	1.9444	1.9148
coverage, N					
100%, 3	0.1901	0.3435	**0.0225**	0.5653	0.5373
100%, 5	0.1181	0.3491	**0.0215**	0.5281	0.5143
100%, 9	0.1212	0.3522	**0.0191**	0.3905	0.3815
100%, 15	**0.1306**	0.3530	–	0.3616	0.3610
50%, 3	0.2066	0.3645	**0.0302**	0.5796	0.5475
50%, 5	0.1302	0.3683	**0.0281**	0.5516	0.5237
50%, 9	0.1356	0.3714	**0.0245**	0.4390	0.4233
50%, 15	0.1430	0.3710	**0.0233**	0.4313	0.4169
25%, 3	0.2429	0.4070	**0.0415**	0.5855	0.5548
25%, 5	0.1719	0.4110	**0.0378**	0.5622	0.5278
25%, 9	0.1734	0.4161	**0.0327**	0.4839	0.4698
25%, 15	0.1820	0.4161	**0.0309**	0.4892	0.4709
10%, 3	0.6604	1.0307	**0.2734**	0.7773	0.7521
10%, 5	0.6264	1.0176	**0.2471**	0.7664	0.7409
10%, 9	0.6230	1.0268	**0.2432**	0.7496	0.7277
10%, 15	0.6283	1.0188	**0.2562**	0.7451	0.7247

Post-processing readouts on KS

The readouts cannot repair forecasts this far off. Over 50 frames the single decode (MSE 0.48 / 0.39) is worse than persistence (0.29); the U-Net is at 5×10^{-8}. On KS the symmetries play the role of the transonic mirror average: averaging the decodes of the 4 shifted × 2 reflected encodings lowers MSE by a factor of 0.54–0.65 (short) and

Table 4: Long-baseline assimilation (`da_test_long`; $\delta_f = 0.1$, $N = 5$, 1000 iterations, 24 problems). U-Net and SDA are capped by cost.

δ_l	T_L	Paper KAE	U-Net	SDA	Ref.	back001
2.5	0.11	0.0182	0.1964	**0.0087**	0.0256	0.0241
10	0.44	0.0300	0.1495	**0.0113**	0.0362	0.0369
25	1.11	0.0388	0.3665	**0.0120**	0.0555	0.0544
50	2.22	0.0402	–	**0.0133**	0.0674	0.1066
100	4.43	**0.0492**	–	–	0.0855	0.0777
200	8.86	**0.0601**	–	–	0.0968	0.0865
400	17.73	**0.0731**	–	–	0.0878	0.1334

Table 5: Forecast from each method's own analysis (canonical window, 24 problems, 2000 iterations).

	analysis	$0.1\,T_L$	$1\,T_L$	horizon (T_L)
U-Net	0.1472	0.0001	0.0027	10.88
Paper KAE	0.0134	0.0141	1.3222	0.53
Reference	0.0365	0.1149	1.6776	0.34
back001	0.0368	0.0997	1.7644	0.40

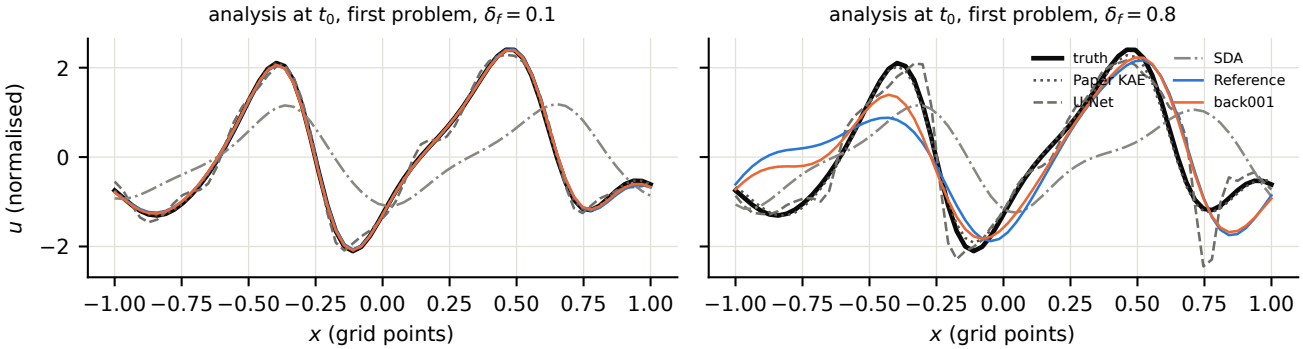


Figure 2: Analyses at t_0 of the first problem of the δ_f sweep, against the truth.

Table 6: The transonic readouts on the CT-KAE's KS forecasts (`test.nc`, from the true state; short: 50 frames from 10 starts per trajectory, long: 400 frames from 3). MSE (normalised) of the single decode, then MSE relative to it (mean ± SE over 3 seeds), and the $\log_{10}$ power ratio prediction / truth at low ($k \leq 3$) and high ($k = 7$–10) wavenumbers. KS is symmetric under periodic shifts and $u(x) \to -u(-x)$; both give members.

		short			long		
		MSE	low	high	MSE	low	high
Reference	single decode	0.482	-0.05	-0.26	2.732	+0.30	+0.98
	+ mirror	0.66±0.13	-0.21	-0.38	0.57±0.11	-0.34	+0.62
	+ 4 shifts × mirror	0.54±0.08	-0.29	-0.52	0.34±0.03	-1.15	-0.29
	+ sharpen	0.50±0.06	-0.09	-0.27	0.44±0.05	-0.43	+0.60
	+ LPM	0.60±0.05	-0.21	+1.67	0.91±0.01	-0.56	+2.51
back001	single decode	0.394	-0.12	-0.32	2.523	+0.19	+0.90
	+ mirror	0.73±0.13	-0.22	-0.42	0.51±0.07	-0.50	+0.46
	+ 4 shifts × mirror	0.65±0.11	-0.27	-0.52	0.37±0.04	-1.12	-0.52
	+ sharpen	0.61±0.09	-0.15	-0.33	0.48±0.08	-0.45	+0.49
	+ LPM	0.69±0.08	-0.22	+1.47	0.92±0.02	-0.59	+2.24
U-Net		4.9e-08			5.4e-05		
persistence		0.291			1.181		

0.34–0.37 (long). But at long leads it removes most of the low-wavenumber power ($\log_{10}$ ratio -1.1): the drop comes from regressing towards the climatological mean, which is blurring. Sharpening keeps most of the gain at short leads (0.50–0.61) and restores power at low k (-0.4), with too much power at high k at long leads. LPM, the best transonic readout, fails here: with 64 points a 16-point patch is a quarter of the domain, and the patch-wise rank matching adds 30–300$\times$ power at $k = 7$–10. On KS, the sharpened symmetry average is the only admissible readout; it brings the 50-frame error (0.24 for both candidates) just under persistence.